Introduction to Kali Linux

Cyber security

Timotej Rozina,
October 2026

Working with Kali Linux

Starting the Kali Linux environment

Tool Name	Menu Category	Core Functionality & Use Case
Wireshark	Sniffing & Spoofing	Network packet analyzer used for network troubleshooting, analysis, and real-time packet capture.
Burp Suite	Web Application Analysis	Integrated platform for performing security testing of web applications, including proxying and vulnerability scanning.
Metasploit Framework	Exploitation Tools	Penetration testing platform that enables finding, exploiting, and validating security vulnerabilities.
John the Ripper	Password Attacks	Fast password cracking tool designed to detect weak passwords using dictionary and brute-force attacks.
Aircrack-ng	Wireless Attacks	Complete suite of tools to assess Wi-Fi network security, focusing on monitoring, attacking, testing, and cracking.

Basic Command Line Commands

Example commands:

```
┌──(kali㉿kali)-[~]
└─$ whoami
hostnamectl
uname -a
df -h
ip a
wget
https://gist.githubusercontent.com/EdwardRayl/3436572afde8ce9e3faf5b7b95356a49/raw/6b25895fce480713560829dec31ac8220ffe5272/gists.txt
sudo apt install 7zip
which nmap
which john
cd /
ls -la
```

```
kali
 Static hostname: kali
       Icon name: computer-vm
         Chassis: vm 💽
      Machine ID: 246bd9421a9a4c0d9da68d5f5abcfabc
         Boot ID: 59e385a33011433aa316accf8c3130ef
  Virtualization: oracle
Operating System: Kali GNU/Linux Rolling
          Kernel: Linux 6.19.14+kali-amd64
    Architecture: x86-64
 Hardware Vendor: innotek GmbH
  Hardware Model: VirtualBox
Hardware Version: 1.2
Firmware Version: VirtualBox
   Firmware Date: Fri 2006-12-01
    Firmware Age: 19y 10month 2w 4d
Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali
6.19.14-1+kali1 (2026-05-05) x86_64 GNU/Linux
Filesystem      Size  Used Avail Use% Mounted on
udev            3.8G     0  3.8G   0% /dev
tmpfs           795M  944K  794M   1% /run
/dev/sda1        79G   17G   59G  22% /
tmpfs           3.9G  4.0K  3.9G   1% /dev/shm
none            1.0M     0  1.0M   0%
/run/credentials/systemd-journald.service
tmpfs           3.9G  200K  3.9G   1% /tmp
none            1.0M     0  1.0M   0%
/run/credentials/getty@tty1.service
tmpfs           795M  108K  795M   1% /run/user/1000
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state
UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel
state UP group default qlen 1000
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.73/24 brd 192.168.1.255 scope global dynamic
noprefixroute eth0
       valid_lft 84570sec preferred_lft 84570sec
```

```
    inet6 2a00:ee2:506:b200:3ac9:6e7a:6cae:13ae/64 scope global
dynamic noprefixroute
       valid_lft 705sec preferred_lft 705sec
    inet6 fdeb:94cb:a2dd:9341:46cb:3be6:be81:c762/64 scope global
dynamic noprefixroute
       valid_lft 1719sec preferred_lft 1719sec
    inet6 fe80::1463:4eb8:89f0:eb97/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
--2026-10-19 10:47:48--
https://gist.githubusercontent.com/EdwardRayl/3436572afde8ce9e3faf
5b7b95356a49/raw/6b25895fce480713560829dec31ac8220ffe5272/gists.tx
t
Resolving gist.githubusercontent.com
(gist.githubusercontent.com)... 185.199.108.133, 185.199.111.133,
185.199.110.133, ...
Connecting to gist.githubusercontent.com
(gist.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9634 (9.4K) [text/plain]
Saving to: 'gists.txt'

gists.txt           100%[================>]   9.41K  --.-KB/s
in 0s

2026-10-19 10:47:49 (17.5 TB/s) - 'gists.txt' saved [9634/9634]

[sudo] password for kali:
The following package was automatically installed and is no longer
required:
  python3-mitmproxy-wireguard
Use 'sudo apt autoremove' to remove it.

Upgrading:
  7zip

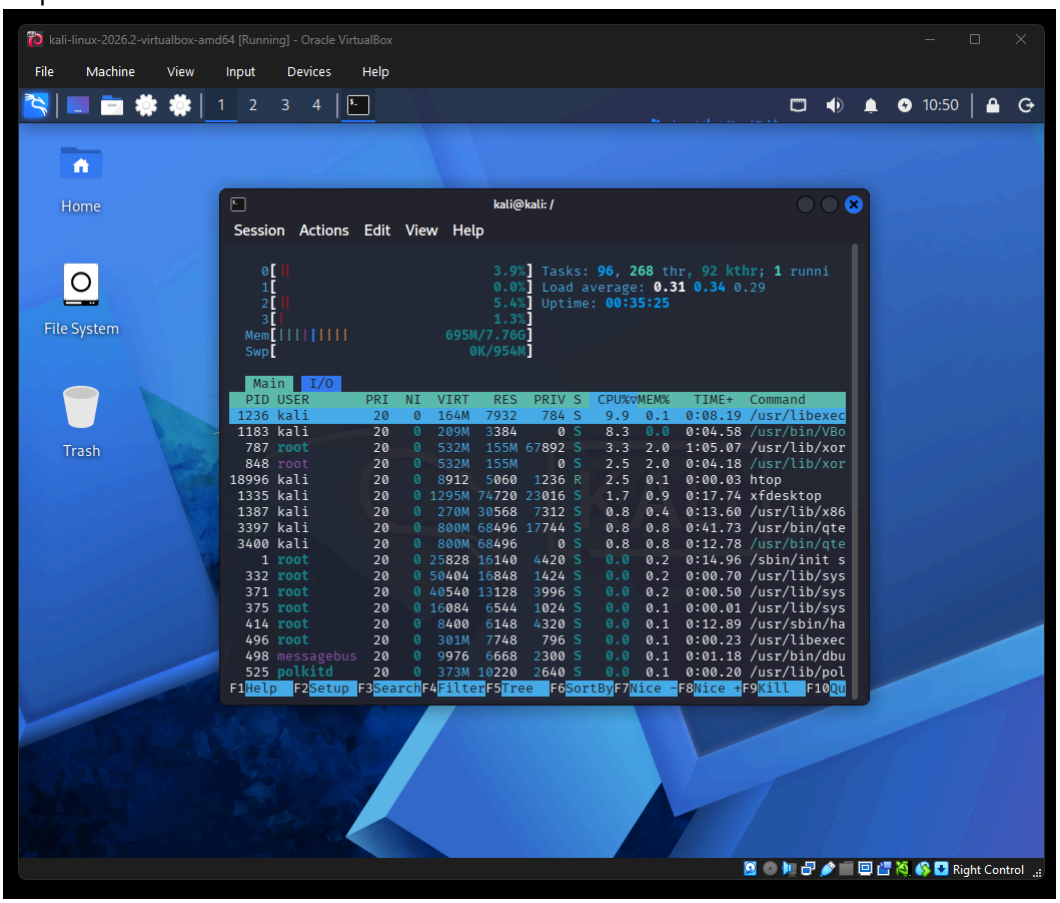
Summary:
  Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 1503
  Download size: 1,717 kB
  Space needed: 411 kB / 62.4 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 7zip amd64
26.03+dfsg-3 [1,717 kB]
```

```
Fetched 1,717 kB in 1s (3,067 kB/s)
(Reading database... 425755 files and directories currently
installed.)
Preparing to unpack .../7zip_26.03+dfsg-3_amd64.deb...
Unpacking 7zip (26.03+dfsg-3) over (26.01+dfsg-2)...
Setting up 7zip (26.03+dfsg-3)...
Processing triggers for kali-menu (2026.2.6)...
Processing triggers for man-db (2.13.1-1)...
/usr/bin/nmap
/usr/sbin/john
total 976644
drwxr-xr-x  18 root root       4096 Oct 19 10:29 .
drwxr-xr-x  18 root root       4096 Oct 19 10:29 ..
lrwxrwxrwx   1 root root          7 Jun 16 06:14 bin -> usr/bin
drwxr-xr-x   3 root root       4096 Jun 16 10:56 boot
drwxr-xr-x  18 root root       3240 Oct 19 10:17 dev
drwxr-xr-x 184 root root      12288 Oct 19 10:17 etc
drwxr-xr-x   3 root root       4096 Jun 16 10:30 home
lrwxrwxrwx   1 root root         34 Jun 16 10:56 initrd.img ->
boot/initrd.img-6.19.14+kali-amd64
lrwxrwxrwx   1 root root         34 Jun 16 10:56 initrd.img.old ->
boot/initrd.img-6.19.14+kali-amd64
lrwxrwxrwx   1 root root          7 Jun 16 06:14 lib -> usr/lib
lrwxrwxrwx   1 root root          9 Jun 16 10:27 lib32 ->
usr/lib32
lrwxrwxrwx   1 root root          9 Jun 16 06:14 lib64 ->
usr/lib64
drwx------   2 root root      16384 Jun 16 10:55 lost+found
drwxr-xr-x   2 root root       4096 Jun 16 10:23 media
drwxr-xr-x   2 root root       4096 Jun 16 10:23 mnt
drwxr-xr-x   3 root root       4096 Jun 16 10:27 opt
dr-xr-xr-x 252 root root          0 Oct 19 10:15 proc
drwx------   6 root root       4096 Oct 19 10:33 root
drwxr-xr-x  41 root root        960 Oct 19 10:17 run
lrwxrwxrwx   1 root root          8 Jun 16 06:14 sbin -> usr/sbin
drwxr-xr-x   3 root root       4096 Jun 16 10:29 srv
-rw-------   1 root root 1000000000 Jun 16 10:56 swap
dr-xr-xr-x  13 root root          0 Oct 19 10:15 sys
-rw-r--r--   1 root root         15 Oct 19 10:28 test
drwxrwxrwt  12 root root        300 Oct 19 10:47 tmp
drwxr-xr-x  15 root root       4096 Jun 16 10:27 usr
drwxr-xr-x  12 root root       4096 Oct 16 07:31 var
```

```
lrwxrwxrwx   1 root root         31 Jun 16 10:56 vmlinuz ->
boot/vmlinuz-6.19.14+kali-amd64
lrwxrwxrwx   1 root root         31 Jun 16 10:56 vmlinuz.old ->
boot/vmlinuz-6.19.14+kali-amd64
```

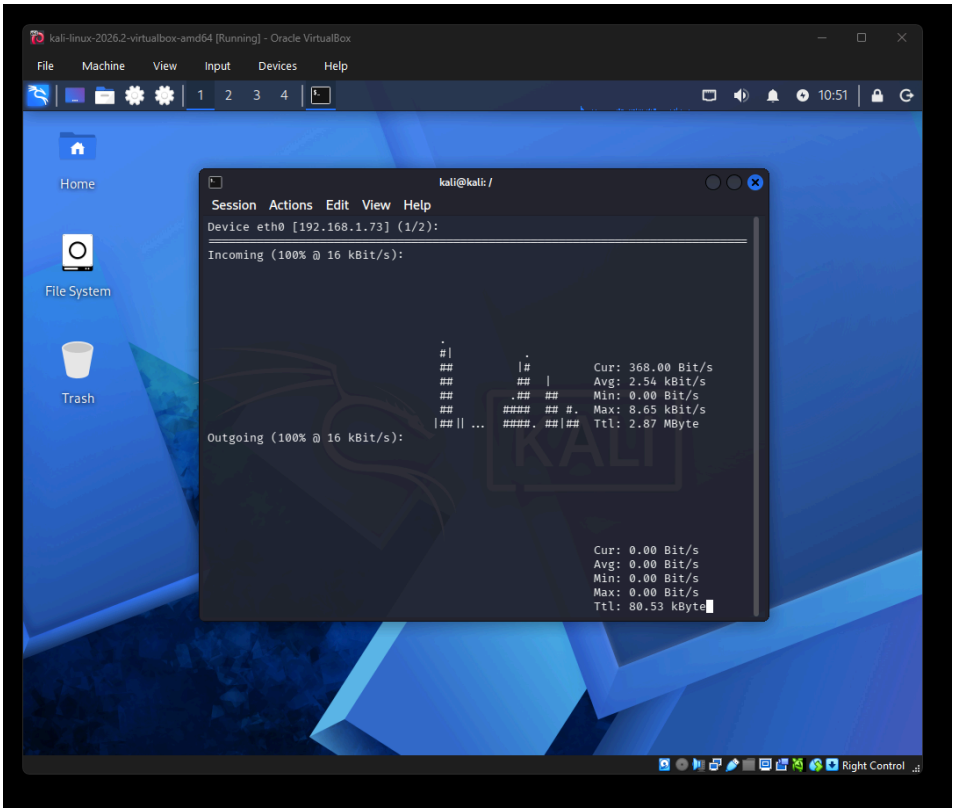
Htop:



traceroute:

```
┌──(kali㉿kali)-[/]
└─$ traceroute google.com
traceroute to google.com (2a00:1450:4025:1803::8b), 30 hops max,
80 byte packets
 1  2a00:ee2:506:b200::1 (2a00:ee2:506:b200::1)  2.376 ms  2.803
ms  3.202 ms
 2  * * *
 3  2a00:ee0:1:10::2 (2a00:ee0:1:10::2)  7.954 ms 2a00:ee0:1:15::2
(2a00:ee0:1:15::2)  7.659 ms  7.464 ms
 4  2a00:ee0:5:17::2 (2a00:ee0:5:17::2)  14.824 ms  15.156 ms
15.344 ms
 5  cv-in-f139.1e100.net (2a00:1450:4025:1803::8b)  16.660 ms
15.852 ms  16.867 ms
```

Nload:



Strings:

```
┌──(kali㉿kali)-[/]
└─$ strings /bin/ls | head
t/lib64/ld-linux-x86-64.so.2
_ITM_deregisterTMCloneTable
__gmon_start__
_ITM_registerTMCloneTable
fgetfilecon_raw
fgetfilecon
freecon
lgetfilecon
lgetfilecon_raw
optind
```

Using tools in Kali Linux

speedtest:

```
┌──(kali㉿kali)-[~/.local/bin]
└─$ speedtest-cli --secure

Retrieving speedtest.net configuration...
Testing from Telekom Slovenije (111.111.111.111)...
Retrieving speedtest.net server list...
Selecting best server based on ping...
Hosted by INTERKA Sp. J. (Olesno) [612.70 km]: 123.272 ms
Testing download
speed.....................................................
...................
Download: 47.85 Mbit/s
Testing upload
speed.....................................................
.....................................
Upload: 18.31 Mbit/s
```

Tcpdump:

```
┌──(kali㉿kali)-[~/.local/bin]
└─$ sudo tcpdump -c 10
[sudo] password for kali:
tcpdump: verbose output suppressed, use -v[v]... for full protocol
decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length
262144 bytes
10:56:34.527578 IP 192.168.1.69.40334 > 255.255.255.255.29810:
UDP, length 847
10:56:34.522769 IP 192.168.1.73.38980 > dns.google.domain: 63006+
PTR? 255.255.255.255.in-addr.arpa. (46)
10:56:34.547075 IP dns.google.domain > 192.168.1.73.38980: 63006
NXDomain 0/1/0 (114)
10:56:34.542759 IP 192.168.1.73.57331 > dns.google.domain: 31249+
PTR? 69.1.168.192.in-addr.arpa. (43)
10:56:34.557023 IP dns.google.domain > 192.168.1.73.57331: 31249
NXDomain 0/0/0 (43)
10:56:34.686687 IP 192.168.1.73.50259 > dns.google.domain: 27016+
PTR? 8.8.8.8.in-addr.arpa. (38)
10:56:34.781194 IP dns.google.domain > 192.168.1.73.50259: 27016
1/0/0 PTR dns.google. (62)
10:56:34.686687 IP 192.168.1.73.34727 > dns.google.domain: 27016+
```

```
PTR? 73.1.168.192.in-addr.arpa. (43)
10:56:34.792385 IP dns.google.domain > 192.168.1.73.34727: 27016
NXDomain 0/0/0 (43)
10:56:35.473100 IP 192.168.1.79.49282
```

Nmap:

```
┌──(kali㉿kali)-[~/.local/bin]
└─$ nmap -O 192.168.1.33
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-19 10:57 -0400
Nmap scan report for 192.168.1.33
Host is up (0.0070s latency).
All 1000 scanned ports on 192.168.1.33 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: A0:36:BC:BB:E5:77 (ASUSTek Computer)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 33.85 seconds
```

Searchsploit:

```
──(kali㉿kali)-[~/.local/bin]
└─$ searchsploit wordpress ftp
-----------------------------------------
--------------------------------
 Exploit Title                          |  Path
-----------------------------------------
--------------------------------
WordPress Plugin MiwoFTP 1.0.5 - Arbitrary | php/webapps/36774.txt
WordPress Plugin MiwoFTP 1.0.5 - Arbitrary | php/webapps/36801.txt
WordPress Plugin MiwoFTP 1.0.5 - Cross-Sit | php/webapps/36761.txt
WordPress Plugin MiwoFTP 1.0.5 - Cross-Sit | php/webapps/36763.txt
WordPress Plugin MiwoFTP 1.0.5 - Multiple  | php/webapps/36762.txt
-----------------------------------------
--------------------------------
Shellcodes: No Results
```

Dnsenum:

```
┌──(kali㉿kali)-[~/.local/bin]
└─$ dnsenum google.com
```

```
dnsenum VERSION:1.3.1

-----   google.com   -----


Host's addresses:
__________________

google.com.                              9         IN    A
142.251.27.113
google.com.                              9         IN    A
142.251.27.102
google.com.                              9         IN    A
142.251.27.139
google.com.                              9         IN    A
142.251.27.100
google.com.                              9         IN    A
142.251.27.138
google.com.                              9         IN    A
142.251.27.101


Name Servers:
_____________

ns3.google.com.                          21600     IN    A
216.239.36.10
ns1.google.com.                          21600     IN    A
216.239.32.10
ns2.google.com.                          21600     IN    A
216.239.34.10
ns4.google.com.                          21600     IN    A
216.239.38.10
```

Gives more like mail servers…

Load balancer:

```
┌──(kali㉿kali)-[~/.local/bin]
└─$ lbd google.com

lbd - load balancing detector 0.4 - Checks if a given domain uses
load-balancing.
                                    Written by Stefan Behte
(http://ge.mine.nu)
                                    Proof-of-concept! Might give
false positives.

Checking for DNS-Loadbalancing: FOUND
google.com has address 142.251.27.113
google.com has address 142.251.27.138
google.com has address 142.251.27.102
google.com has address 142.251.27.139
google.com has address 142.251.27.101
google.com has address 142.251.27.100

Checking for HTTP-Loadbalancing [Server]:
 gws
 NOT FOUND

Checking for HTTP-Loadbalancing [Date]: 15:00:46, 15:00:46,
15:00:47, 15:00:47, 15:00:47, 15:00:47, 15:00:48, 15:00:48,
15:00:48, 15:00:48, 15:00:48, 15:00:49, 15:00:49, 15:00:49,
15:00:49, 15:00:49, 15:00:50, 15:00:50, 15:00:50, 15:00:50,
15:00:50, 15:00:51, 15:00:51, 15:00:51, 15:00:51, 15:00:51,
15:00:52, 15:00:52, 15:00:52, 15:00:52, 15:00:52, 15:00:53,
15:00:53, 15:00:53, 15:00:53, 15:00:53, 15:00:54, 15:00:54,
15:00:54, 15:00:54, 15:00:54, 15:00:55, 15:00:55, 15:00:55,
15:00:55, 15:00:55, 15:00:56, 15:00:56, 15:00:56, 15:00:56, NOT
FOUND

Checking for HTTP-Loadbalancing [Diff]: FOUND
< Content-Security-Policy-Report-Only: object-src 'none';base-uri
'self';script-src 'nonce-2-H3BLPWjJ6w5zSfQhEmEg' 'strict-dynamic'
'report-sample' 'unsafe-eval' 'unsafe-inline' https:
http:;report-uri https://csp.withgoogle.com/csp/gws/other-hp
> Content-Security-Policy-Report-Only: object-src 'none';base-uri
'self';script-src 'nonce-NEJruwfgMKlYv3jEXRg6WA' 'strict-dynamic'
'report-sample' 'unsafe-eval' 'unsafe-inline' https:
```

```
http:;report-uri https://csp.withgoogle.com/csp/gws/other-hp

google.com does Load-balancing. Found via Methods: DNS HTTP[Diff]
```

Nth:

```
┌──(kali㉿kali)-[~/.local/bin]
└─$ nth -t 'ef487f75307f96954d3bb132e5f4b035'


  _   _                          _____ _           _          _
_
_
 | \ | |                        |_   _| |         | |        | |
| |
| |
 |  \| | __ _ _ __ ___   ___ ______| | | |__   __ _| |_ _____|
|_| | __ _
___| |__
 | . ` |/ _` | '_ ` _ \ / _ \______| | | '_ \ / _` | __|_____|  _
|/ _` /
__| '_ \
 | |\  | (_| | | | | | |  __/      | | | | | (_| | |_       | |
| | (_| \__
\ | | |
 \_| \_/\__,_|_| |_| |_|\___|      \_/ |_| |_|\__,_|\__|      \_|
|_/\__,_|___/_| |_|

https://twitter.com/bee_sec_san
https://github.com/HashPals/Name-That-Hash


ef487f75307f96954d3bb132e5f4b035

Most Likely
MD5, HC: 0 JtR: raw-md5 Summary: Used for Linux Shadow files.
MD4, HC: 900 JtR: raw-md4
NTLM, HC: 1000 JtR: nt Summary: Often used in Windows Active
Directory.
Domain Cached Credentials, HC: 1100 JtR: mscach

Least Likely
Domain Cached Credentials 2, HC: 2100 JtR: mscach2 Double MD5, HC:
```

```
2600
Tiger-128,  Skein-256(128),  Skein-512(128),  Lotus Notes/Domino
5, HC: 8600
JtR: lotus5 md5(md5(md5($pass))), HC: 3500 Summary: Hashcat mode
is only
supported in hashcat-legacy. md5(uppercase(md5($pass))), HC: 4300
md5(sha1(pass)),HC:4400 md5(utf16(pass)), JtR: dynamic_29
md4(utf16(pass)),JtR:dynamic33md5(md4(pass)), JtR: dynamic_34
Haval-128, JtR: haval-128-4 RIPEMD-128, JtR: ripemd-128 MD2, JtR:
md2
Snefru-128, JtR: snefru-128 DNSSEC(NSEC3), HC: 8300  RAdmin v2.x,
HC: 9900
JtR: radmin Cisco Type 7,  BigCrypt, JtR: bigcrypt
```

Reflection and Analysis

Why do we use Kali Linux? What is the advantage of Kali Linux compared to other Linux distributions? Which features and tools of Kali Linux attracted you the most?

We use kali linux because it provides an out of the box environment designed for offensive security and digital forensics. This greatly decreases the time needed to manually set up the environment. This purpose built environment is its biggest advantage.

The most attractive feature of Kali Linux is that over 600 security tools come pre-installed and neatly organized into categories in the application menu. Instead of wasting time installing software and fixing missing dependencies, everything works right out of the box.